Appendix D. Hyperparameter sensitivity analysis

To examine the robustness of the hyperparameter settings in Table 3, a one-factor-at-a-time sensitivity analysis is conducted. In each test, one hyperparameter is varied across three levels, while the remaining parameters are fixed at their selected values. The first-stage EA is evaluated using Cases 3, 5, and 10, and the second-stage EA is evaluated using a synthetic instance with 132 panels. Each setting is run ten times with different random seeds, and the results are reported in Tables 12 and 13. For the mutation percentage ptMst1, whose selected value is 5/Nsegs (Table 3), the three levels are 0.5, 1, and 2 times this value, i.e., 2.5/Nsegs, 5/Nsegs, and 10/Nsegs.

The sensitivity analysis results show that the selected hyperparameters balance solution quality and computational efficiency. For the first-stage EA (Table 12), the crossover probability, mutation probability, and fitness constant have no significant influence within the tested ranges, whereas the mutation percentage is the most sensitive parameter. Increasing the population size or the number of generations improves solution quality by no more than 2% but nearly doubles the computational cost. For the second-stage EA (Table 13), none of the tested hyperparameters has a statistically significant effect, with all p-values greater than or equal to 0.25. The differences among parameter settings remain within the run-to-run variance of approximately ± 2-5 m, indicating that the results are insensitive to these settings. Overall, the selected hyperparameters are robust and conservative choices.

Table 12

Sensitivity analysis of the first-stage EA hyperparameters

Parameter	Value	Case 3	Case 5	Case 10	Time (*s*)
pbC_{st1}	0.6	5.46±0.20	9.26±0.27	17.07±0.22	1.00
	0.8	**5.44±0.16**	**9.21±0.22**	**17.08±0.23**	**1.00**
	1.0	5.52±0.14	9.31±0.24	17.08±0.22	1.00
pbM_{st1}	0.2	5.53±0.13	9.45±0.15†	17.19±0.25	0.98
	0.4	**5.44±0.16**	**9.21±0.22**	**17.08±0.23**	**1.00**
	0.6	5.47±0.14	9.24±0.28	16.94±0.29	0.99
ptM_{st1}	0.1	5.40±0.17	9.00±0.22	16.67±0.34	0.99
	0.2	**5.44±0.16**	**9.21±0.22**	**17.08±0.23**	**1.00**
	0.4	5.62±0.00	9.60±0.02	17.91±0.15	0.97
$N_{pop,st1}$	8	5.61±0.02	9.37±0.28	17.39±0.31	0.49
	16	**5.44±0.16**	**9.21±0.22**	**17.08±0.23**	**1.00**
	32	5.35±0.18	9.03±0.28	16.73±0.45	1.96
Ng_{st1}	50	5.52±0.14	9.41±0.19	17.32±0.25	0.49
	100	**5.44±0.16**	**9.21±0.22**	**17.08±0.23**	**1.00**
	200	5.37±0.23	9.11±0.28	16.90±0.26	1.97
C_{st1}	100	5.44±0.16	9.24±0.22	17.08±0.23	0.98
	1000	**5.44±0.16**	**9.21±0.22**	**17.08±0.23**	**1.00**
	10000	5.44±0.16	9.21±0.22	17.08±0.23	0.95

Table 13
Sensitivity analysis of the second-stage EA hyperparameters

Parameter	Value	Waste (m)	Best	Time (s)	p
pbC_{st2}	0.05	24.63±5.33	16.79	36.9	0.70
	0.1	**26.24±3.37**	**20.55**	**38.8**	—
	0.2	25.33±4.33	21.08	38.3	0.56
pcM_{st2}	0.025	25.68±3.40	21.68	39.1	1.00
	0.05	**26.24±3.37**	**20.55**	**38.8**	—
	0.1	25.99±3.52	20.99	40.9	1.00
$N_{pop,st2}$	512	24.16±2.48	18.81	20.7	0.38
	1024	**26.24±3.37**	**20.55**	**38.8**	—
	2048	26.25±4.11	19.02	76.3	1.00
$N_{g,st2}$	250	24.81±1.80	21.46	19.7	0.30
	500	**26.24±3.37**	**20.55**	**38.8**	—
	750	26.27±3.96	19.37	57.4	0.25